

AI × City Climate Action Hackathon 2026

Route2Zero

Team Larpers · Transparent corridor intelligence for a just e-jEEPney transition.

A transparent, export-ready decision-support MVP

Team Larpers · METRO MANILA · 2026

1,522

ROUTES SCREENED

GTFS route-direction records

71.4

TOP PRIORITY

Default scenario score

WHY THIS MATTERS

A route ranking without evidence can hard-code the wrong trade-offs.

- **High activity does not automatically mean high readiness.**
 - Equity proxies must be visible—not treated as ground truth.
 - Historic service data cannot stand in for present operations.
 - Operators and riders must be part of validation before investment.
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THE DECISION QUESTION

Which corridors deserve deeper electrification work first?

Route2Zero turns a fragmented evidence base into a shortlist that city teams can inspect, challenge and validate.

- **Screen 1,522 route-direction records consistently.**
 - Balance climate value, equity exposure and feasibility.
 - Export the same decision context into formal reports.
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TARGET USERS AND COMMUNITIES

Built for the institutions and people who shape the transition

- **LTFRB and partner LGUs prioritising corridor validation.**
 - Jeepney cooperatives and drivers navigating finance and operations.
 - Climate and development financiers targeting accountable investment.
 - Transit-dependent riders, including informal and marginalised communities.
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PRODUCT PROMISE

One transparent decision surface—without pretending uncertainty is precision.

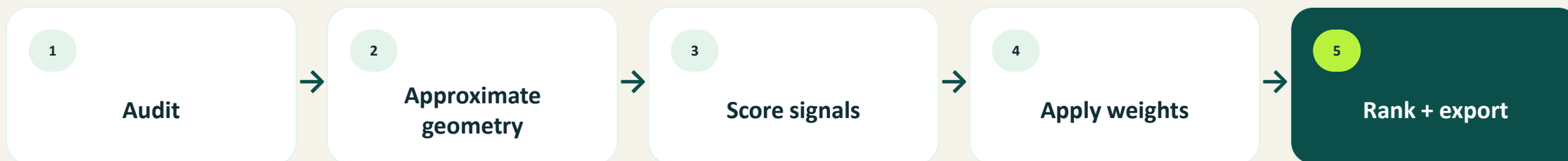
The rank is deterministic. Scenario controls are visible. AI can explain results, but it cannot change a score.

“

Transparent enough to challenge.
Useful enough to act on.

HOW THE PRODUCT WORKS

From source evidence to a challengeable action queue



DEFAULT POLICY LENS

Four interpretable signals, normalised to 100%

35%

Emissions potential

Route activity proxy

35%

Equity density

WorldPop catchment proxy

15%

Grid feasibility

Luzon renewable-share proxy

15%

Operator readiness

Neutral placeholder until verified

AI BOUNDARY

AI explains. The scoring pipeline ranks.

DETERMINISTIC

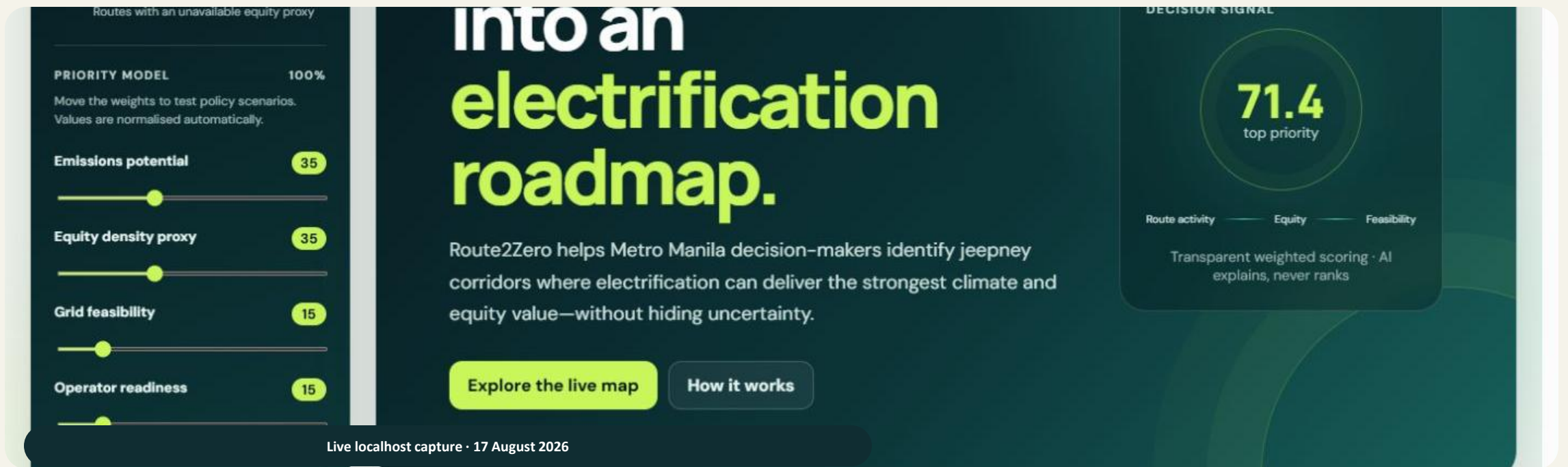
- Activity proxy
- Equity proxy
- Grid proxy
- Operator placeholder
- Weighted sum

OPTIONAL AI

- Plain-language rationale
- Evidence request summary
- Never edits score columns
- Never changes rank order

LIVE DASHBOARD

A 2026 decision cockpit built for judges and city teams



MAPBOX ROAD GEOMETRY

Selected routes follow the street network



SCENARIO TESTING

Weights update the action queue without hiding the policy choice.

- **Move emissions, equity, grid and operator sliders.**
 - Values are normalised automatically.
 - Every export records the current city lens and weights.
 - Reduced-confidence routes remain an explicit opt-in.
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CURRENT ACTION QUEUE

Top routes for deeper feasibility work

Routes with an unavailable equity proxy

PRIORITY MODEL100%

Move the weights to test policy scenarios.
Values are normalised automatically.

Emissions potential35

Equity density proxy35

Grid feasibility15

Operator readiness15

#2	Dasmarinas Resettlement Area – Baclaran via Coastal Rd. LTFRB_PUJ1241	Pasay	69.7	97.5	73.3	16.1	50.0
#3	BACLARAN – DASMARINAS B BAYAN LTFRB_PUJ2083	Dasmarinas	69.5	96.7	73.6	16.1	50.0
#4	Binangonan – Pasig (TP) LTFRB_PUJ1156	Biñan	68.7	97.6	70.3	16.1	50.0
#5	Dasmarinas Resettlement Area – Baclaran via Coastal Rd. LTFRB_PUJ1240	Dasmarinas	68.2	93.8	72.6	16.1	50.0
#6	Francisco Homes – Cubao LTFRB_PUJ1352	Quezon City	68.1	89.7	76.6	16.1	50.0
#7	BACLARAN – DASMARINAS B BAYAN LTFRB_PUJ2084	Parañaque	67.7	91.4	73.6	16.1	50.0
#8	Binangonan – Pasig (TP)	Pasig	67.2	93.6	70.2	16.1	50.0

Top-ranked records under the default 35 / 35 / 15 / 15 policy lens.

LEADING SCREENING RESULT

Francisco Homes – Cubao

71.4

PRIORITY

Rank #1 under defaults

98.2

EMISSIONS

Activity proxy

77.4

EQUITY

Density proxy

16.1

GRID

Regional proxy

Operator readiness = 50.0 neutral placeholder

DECISION PACK

The dashboard exports the live policy context in three formats.

PDF

Print-ready decision report

DOC

Editable Word report

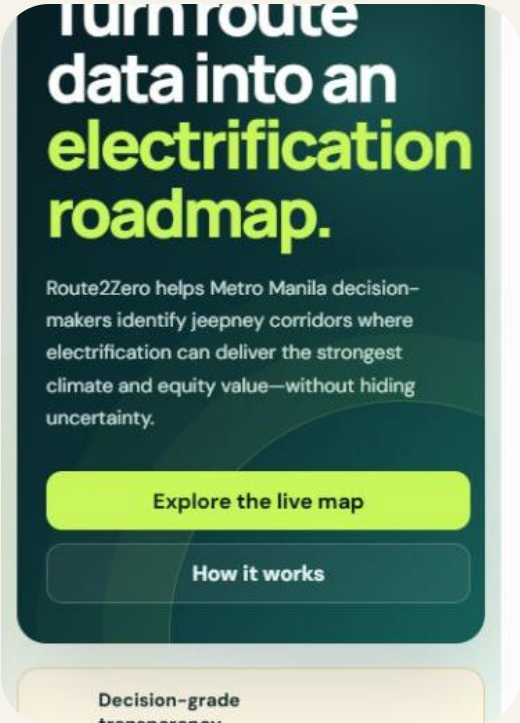
CSV

Scenario ranking data

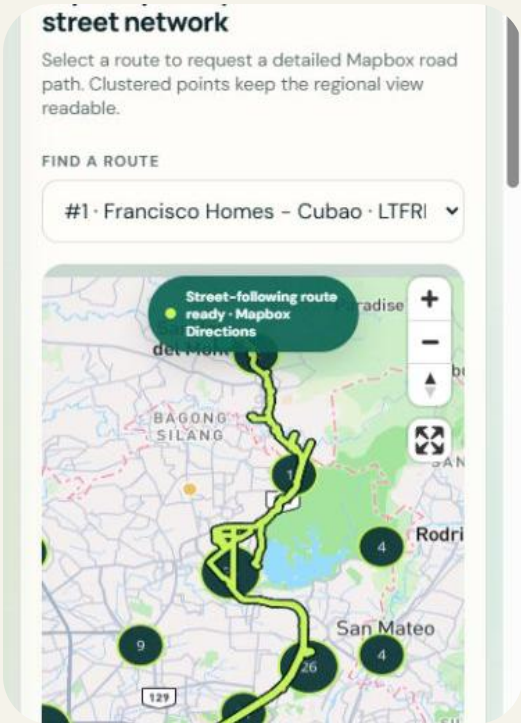
RESPONSIVE ACCESS

The same decision story works on mobile

390 × 844 viewport

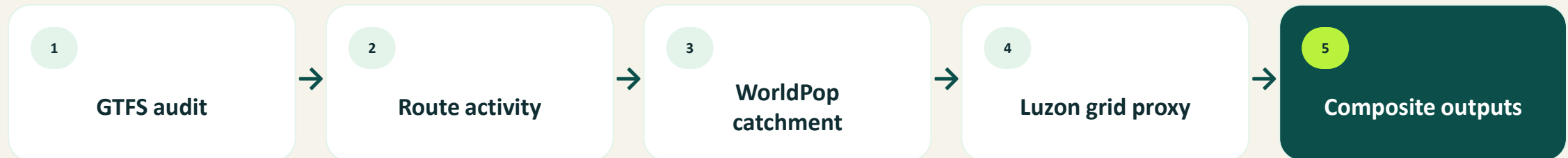


No horizontal overflow



REPRODUCIBLE PIPELINE

Evidence stays traceable from source fields to the interface.



VISIBLE LIMITATIONS

Every important approximation becomes a validation task.

- **GTFS service calendar ends in 2020.**
 - 1,520 route records use stop-sequence geometry approximation.
 - WorldPop is density—not informal-settlement status.
 - Grid feasibility is regional, not depot-specific.
 - Operator readiness defaults to 50 until verified.
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EXPECTED REAL-WORLD BENEFITS

A faster, fairer path from screening to implementation

- **Focus limited validation resources on the most promising corridors.**
 - Protect transit access by making equity trade-offs visible.
 - Target route, depot, grid and finance evidence before procurement.
 - Create a shared, auditable decision record across institutions.
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ACCOUNTABILITY

A multidisciplinary team with named owners

John Marwin Ebona

Product lead

Prince Marl

Data lead

Joaquin Sarmiento

Mobility lead

Isaac Marcus

Engineering lead

Andrei Dela Cruz

Policy lead

Russel Mendez

Research lead

Tj Moreno

Engagement lead

JM Palaganas

QA + documentation

UNIQUE INNOVATION

Route2Zero turns imperfect data into an accountable validation agenda.

Transparent scoring · street-following Mapbox routes · visible uncertainty · export-ready decision packs

SCREEN → VALIDATE → CO-DESIGN → ACT

route2zero · transparent by design